

Augmented Intelligent using Image Processing and LLM: A Case Study of Patient Triage System

977-401 DIGITAL ENGINEERING PROJECT II
SEMESTER 1/2026

ADVISOR: AJ.WISARUT CHANTARA
ASST.PROF. DR.KOMSAN KANJANASIT
ASST.PROF. KULLAWAT CHAOWANAWATEE

PRESENT BY PHOKIN WANNA

PROJECT BACKGROUND

Problems

- Overcrowded emergency rooms and OPD areas cause delays because nurses have overloaded and spend too much time asking repetitive intake questions

Proposed Solution

- Uses computer vision and local LLM to pre-screen patients and categorize urgency using ESI standards and complete patient history interview (Above ESI lvl 2)

DE II Focus

- To detect facial skin anomalies accurately and integrate all modules into a working run time kiosk system concept



SYSTEM ARCHITECTURE

Pipeline Design

- The system is separated into model preparation and run time kiosk execution

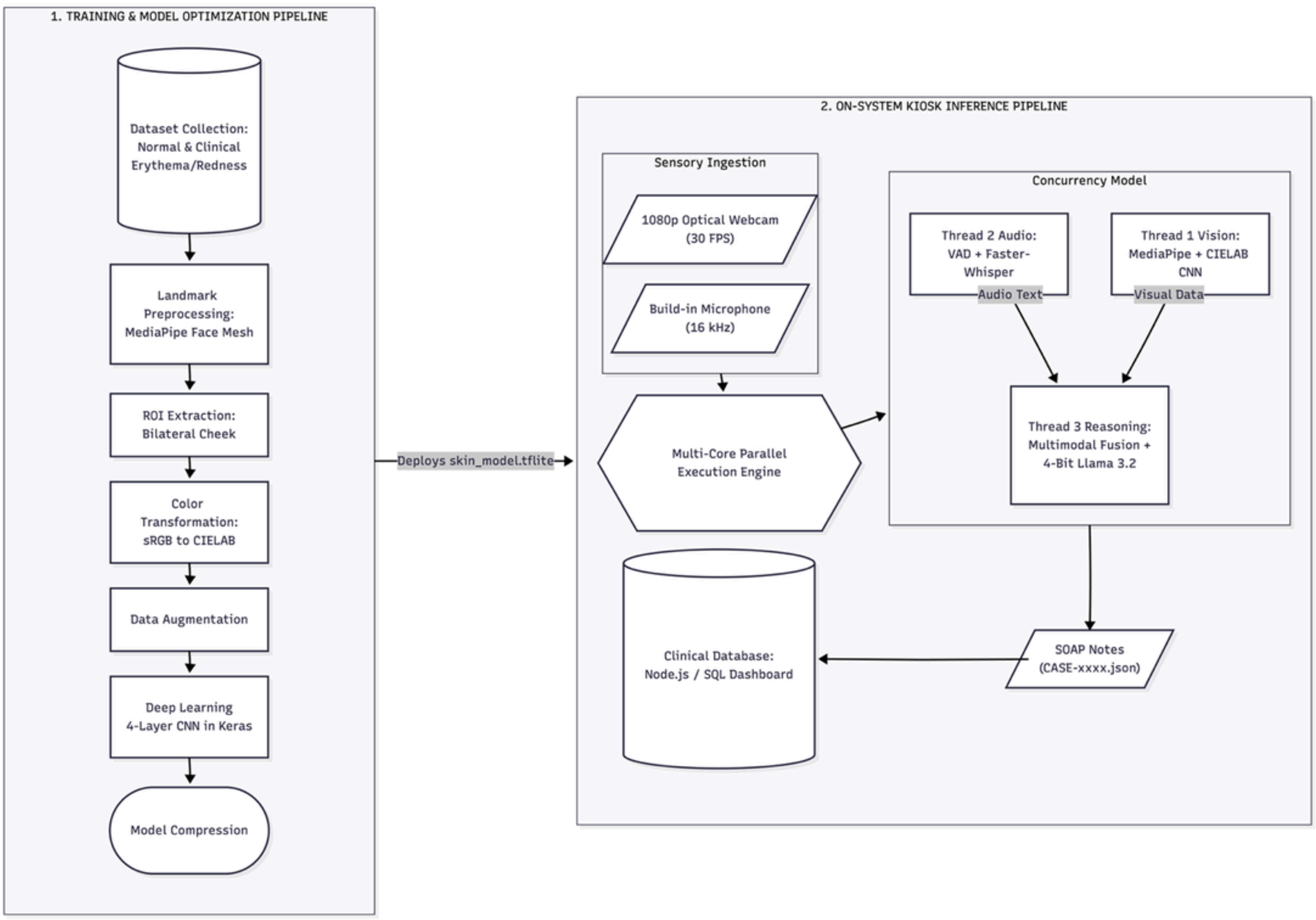
Training Pipeline

- Extract facial skin patches then convert them into the CIELAB color space and train a compact CNN before exporting it to TFLite for edge deployment

Kiosk Inference Pipeline

- The kiosk runs on a local hardware using multi threading to handle camera tracking, speech recognition and a local Llama 3.2 model simultaneously without cloud dependency

SYSTEM ARCHITECTURE (CONT.)



PREVIOUS WORK & LIMITATIONS

Posture Recognition (72.0% Accuracy)

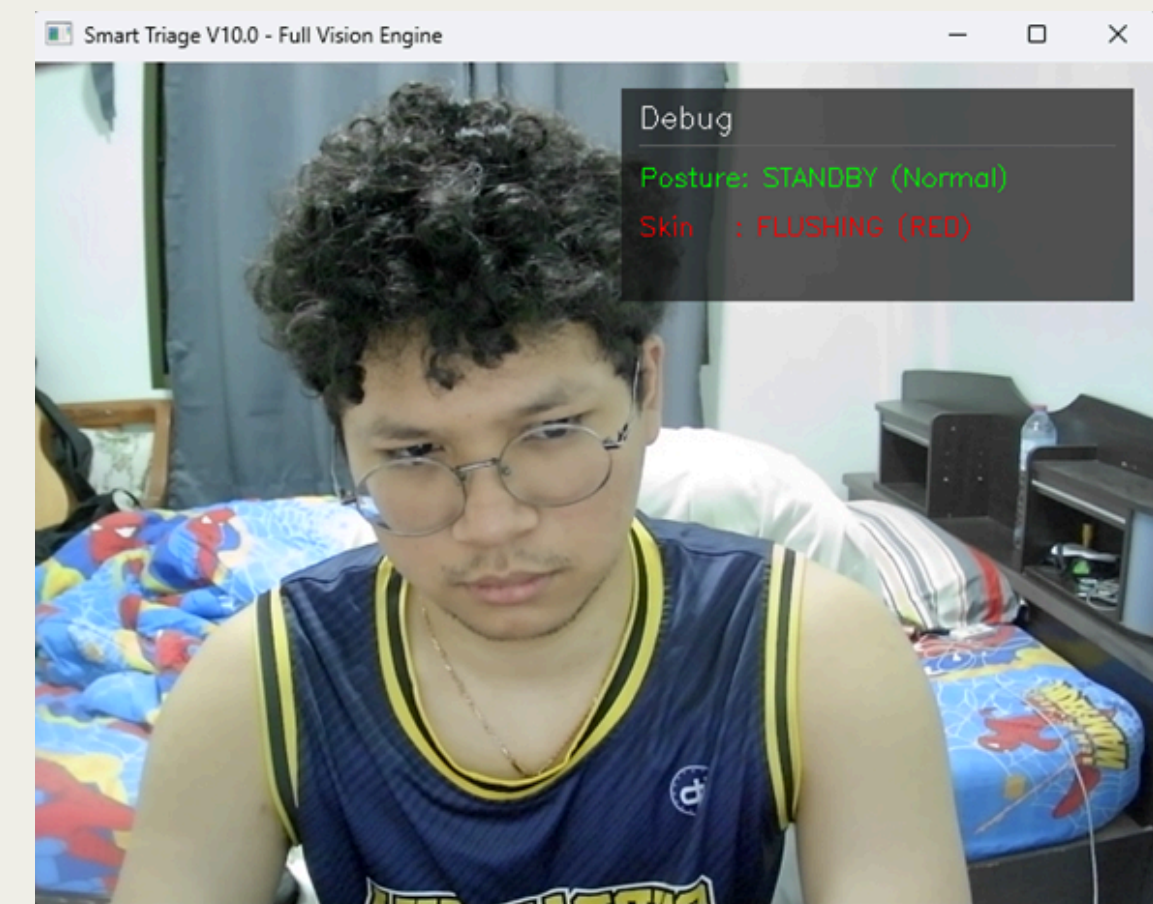
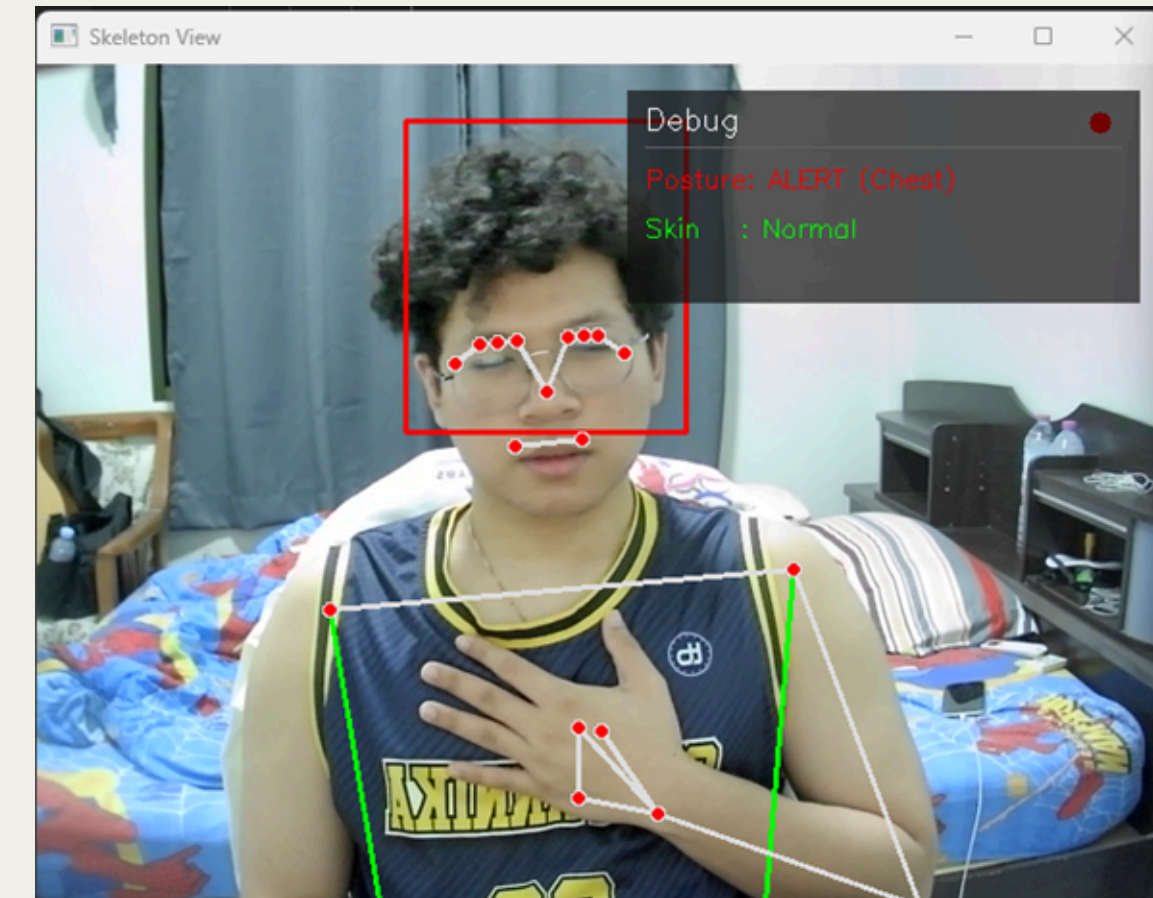
- Successfully detected pain gestures such as chest and stomach clutching, but off axis body turns sometimes caused errors

Skin Color Analysis (56.0% Accuracy)

- The old manual HSV method failed because monitor glare directly reflected on the patient's face and caused false pallor alerts

System Latency

- Running all tasks sequentially without multi threading caused the camera to freeze whenever audio or AI reasoning was executing.



PREVIOUS WORK & LIMITATIONS (CONT.)

<i>Subsystem Component</i>	<i>(Project DE I) Previous Baseline</i>	<i>Root Cause</i>	<i>Work Improvement</i>
Skin Color Analysis	56.0% Accuracy	Light and color mix up too much in HSV and nearby lights cause too many issues.	Switch to CIELAB and replace manual rules with CNN models trained on 20,000 skin samples.
Gesture Classification	72.0% Accuracy	The 2D view gets confusing when bodies turn or key tracking points are blocked.	Use more data to improve tracking over time and support multiple angles.
Clinical Reasoning & Output	12.8 - 15.0s Latency	Processing fixed prompts takes too long and the text varies too much.	Use standard key-value inputs, set clear rules for ESI level prompts and ensure JSON outputs follow PDPA laws.
Clinical Interfacing	Command-line logs / Basic GUI	Nurses do not have a central screen to see all visual data.	Launch a web dashboard for nurses to show ESI urgency, SOAP medical summaries and proof photos.

CURRENT WORK: FACE EXTRACTION & POSE LOGIC

Skin Segmentation

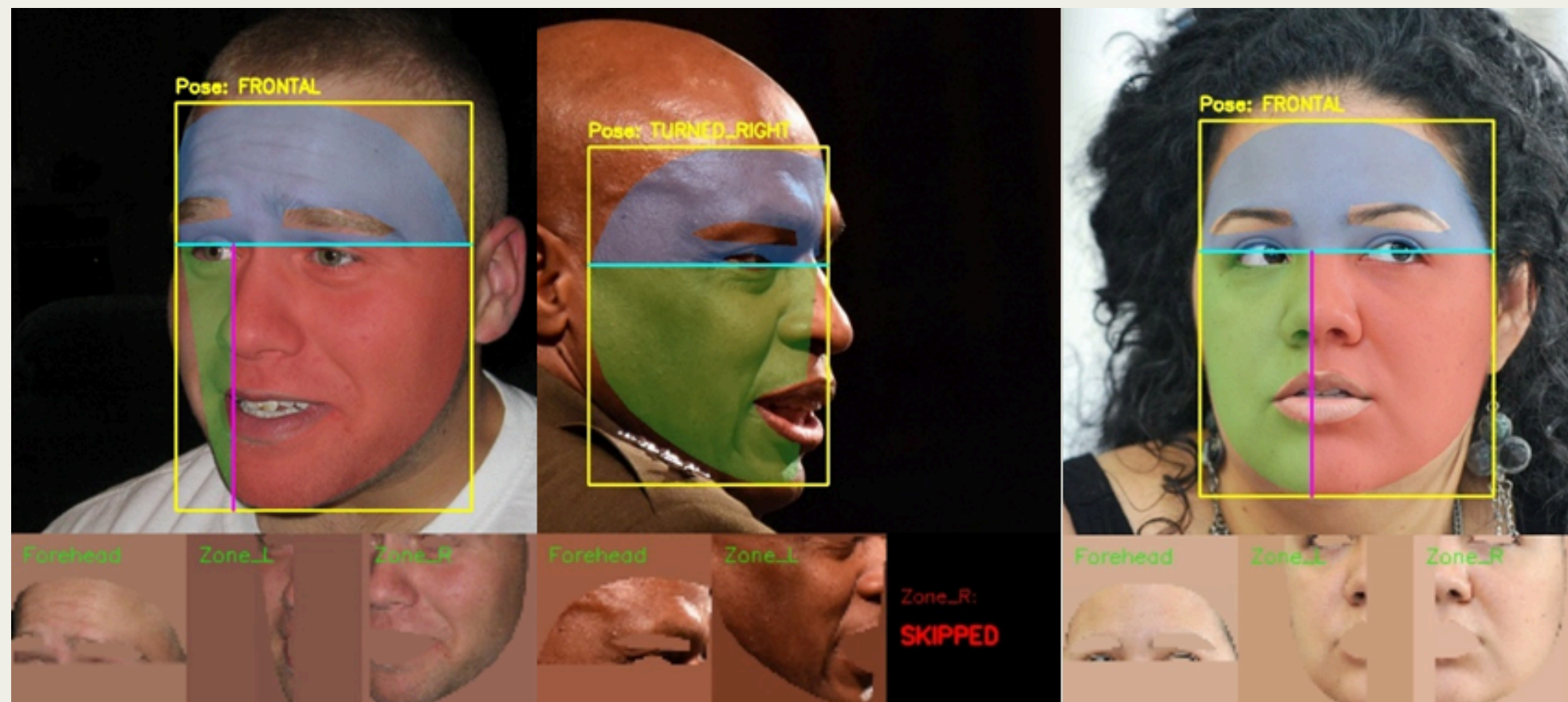
- Instead of checking only one cheek point, the system divides the face into three clear zones: Forehead, Left Cheek and Right Cheek.

Non Skin Masking

- Facial landmarks automatically mask out the eyes, mouth and hair so that only pure skin pixels are sent to the NNs model.

Pose Adaptation

- When patients turn their head, the system detects the angle and automatically marks the hidden cheek as "SKIPPED" to prevent shadows from creating false alarms.



CURRENT WORK: CIELAB SKIN WITH CNN MODELS

<i>Metric / Parameter</i>	<i>Baseline Prototype (DE I HSV)</i>	<i>Model 01: Cross Entropy Loss (CEL)</i>	<i>Model 02: Binary Focal Loss (FL)</i>
Input Representation	Heuristic HSV Threshold	CIELAB Patches	CIELAB Patches
Evaluation Dataset	70 Test Trials	Patient-Level Test Set	Validation Set
Overall Accuracy	56.00%	79.90%	83.75%
Precision: Flushing	52.40%	0.52	0.9821
Precision: Normal	-	0.96	-
Recall / Sensitivity: Flushing	58.30%	0.883	0.2644
Recall / Specificity: Normal	54.00%	0.776	0.9986
F1-Score: Flushing	0.551	0.66	0.4167
F1-Score: Normal	-	0.86	-
Area Under ROC (ROC-AUC)	N/A	0.9142	0.9738
Area Under PR (PR-AUC)	N/A	-	0.9203
Native Model Size (Keras)	N/A	0.11 MB	0.11 MB
Quantized Edge Size (TFLite)	N/A	0.03 MB (30 KB)	0.03 MB (30 KB)

Model 01 (Cross Entropy Loss)

- It work as a high sensitivity screener by catching 88.3% of flushing cases, but it still produces some false alarms (52.0% precision)

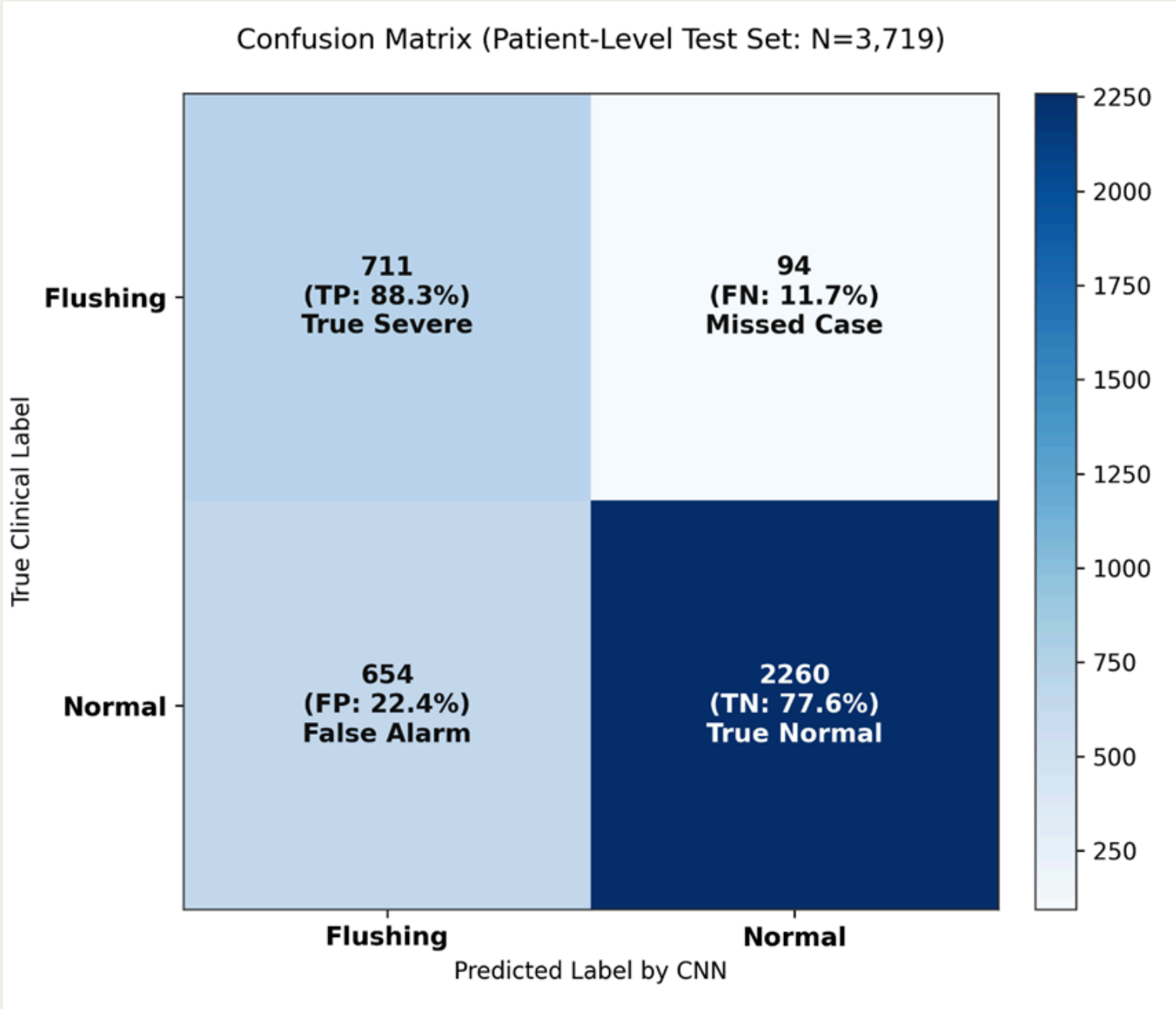
Model 02 (Focal Loss)

- Solves the false alarm problem completely with 99.86% specificity and only 4 false positives out of 2,959 normal samples

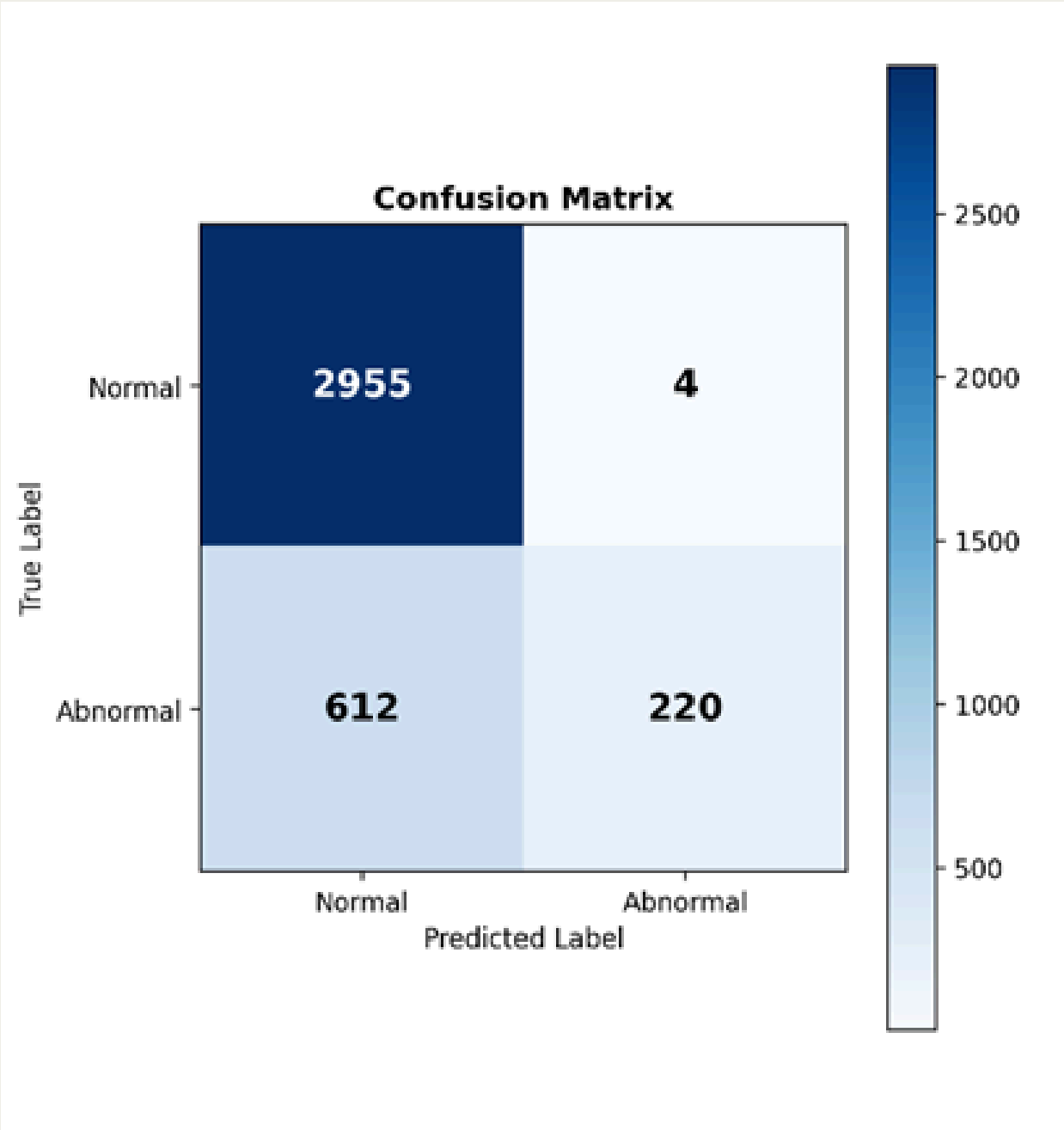
CIELAB Color Space

- Switched from HSV to CIELAB because it separates brightness from redness, which protects the system from lighting reflections

CURRENT WORK: CIELAB SKIN WITH CNN MODELS (CONT.)

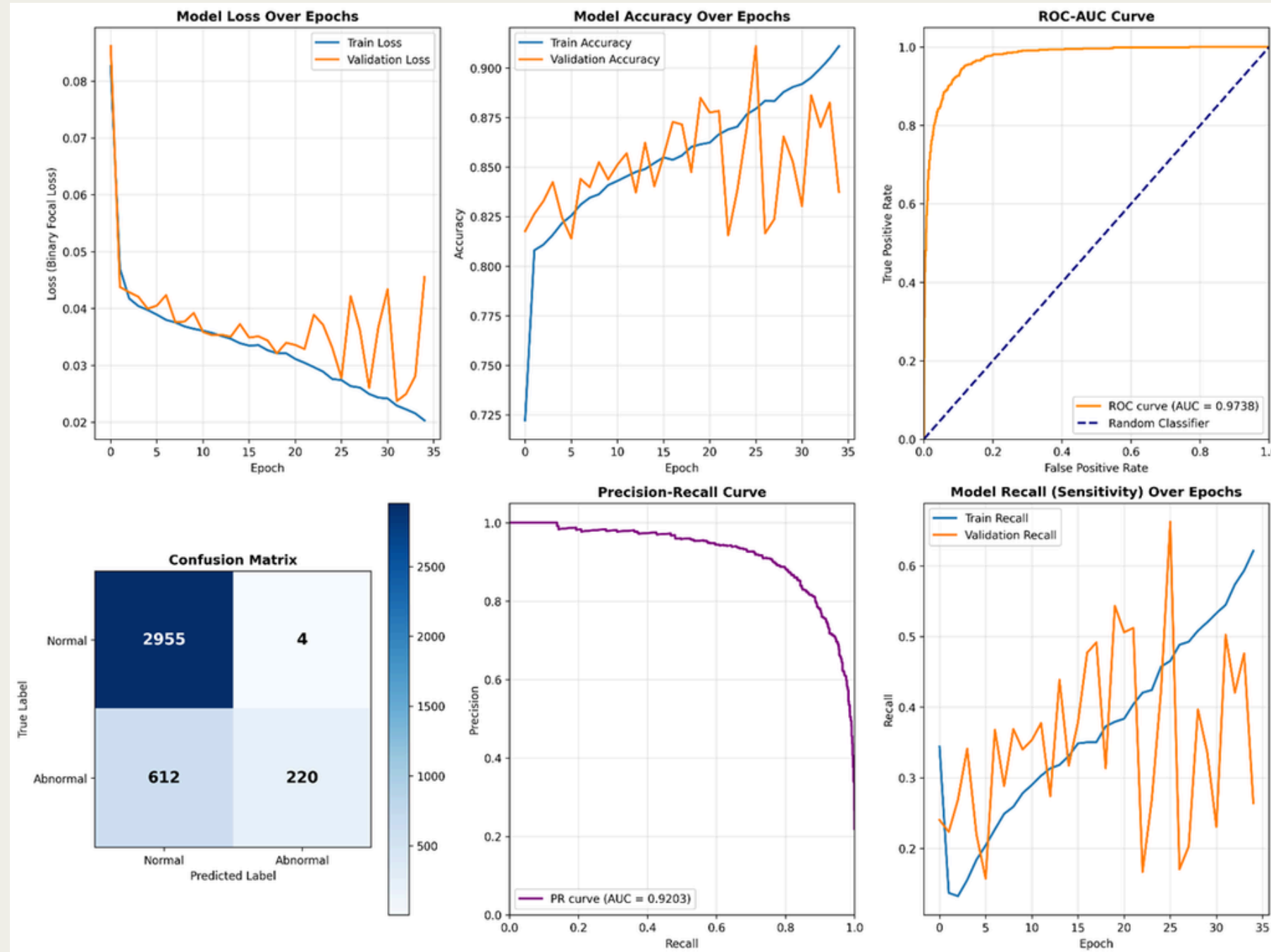


MODEL 01 (CEL)



MODEL 02 (FL)

MODEL 02 ANALYSIS & IMPROVEMENT



Identified Flaws

- The model began overfitting because validation loss started trending upward while loss kept dropping

Low Sensitivity

- At the default 0.50 cutoff, the model missed 612 abnormal cases out of 832, resulting in a low recall of only 26.44%

High Inherent Separation (AUC 0.9738)

- The very high ROC-AUC and 99.86% specificity prove the CNN learned strong features, just threshold was set too strict

CURRENT WORK: GESTURE & POSTURE RECOGNITION RESULTS

Dynamic Torso Boundary (0.55 Ratio)

- Setting the chest boundary at 55% of torso length successfully separates chest pain from stomach pain even when patients bend over

2.0 Seconds Confirmation

- Postures must be held for 60 consecutive frames to trigger an alert so that quick accidental movements are ignored

False Positive Filters

- Head Scratching: suppressed using Z-depth behind the ears
- Crossed Arms: suppressed using forearm vector intersection
- Thinking Pose: suppressed using hand to mouth distance

CURRENT WORK: PARALLEL PROCESSING

The Problem with Single Thread

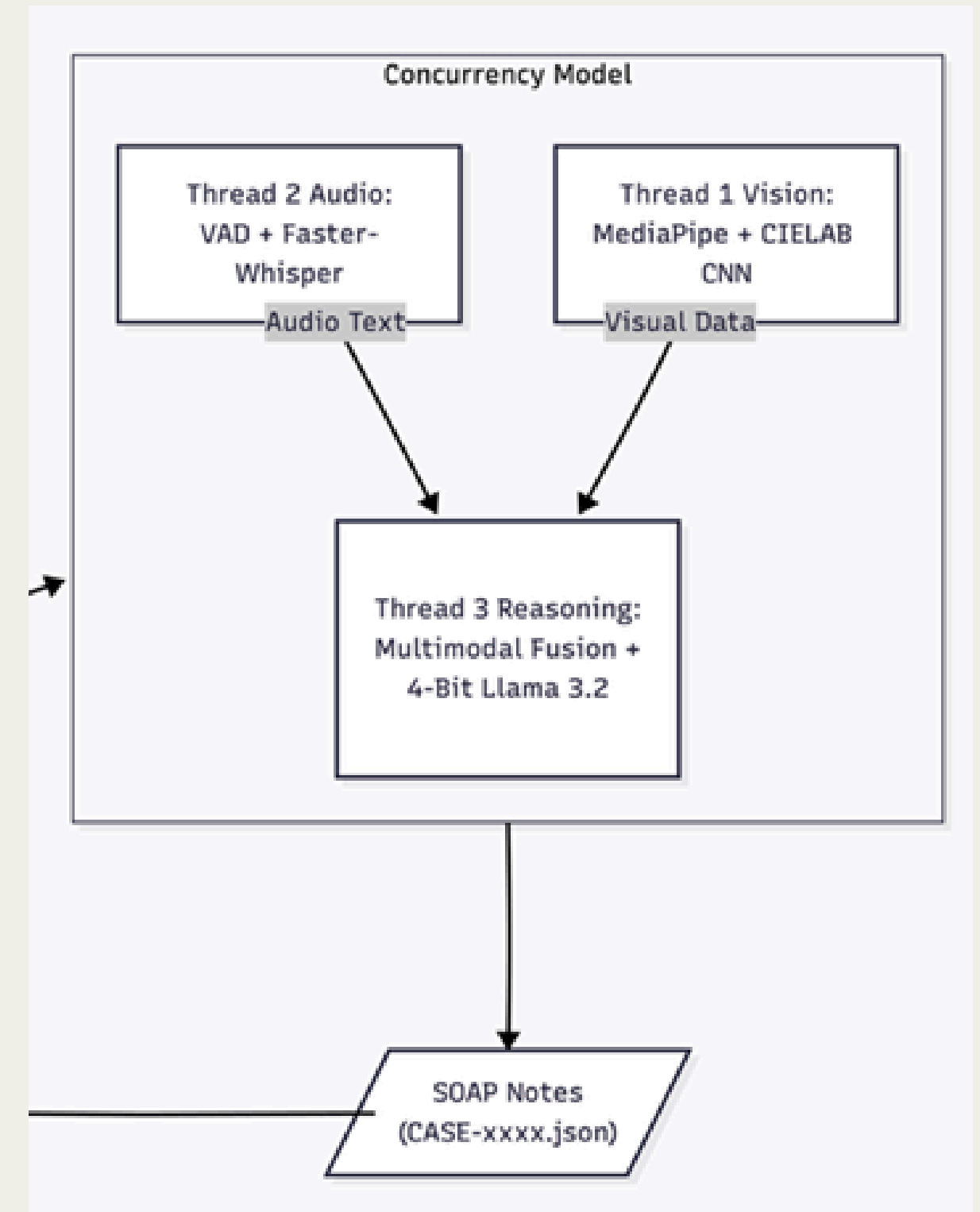
- Running all code in a single loop caused the camera to freeze at 0 FPS whenever the LLM was thinking, stretching the total cycle time to 12.8–15.0 seconds

Asynchronous Multi Thread Pipeline

- Vision Thread keeps camera tracking smooth at 30 FPS.
- Audio Thread continuously listens for patient speech.
- Reasoning Thread processes the LLM in the background.

Performance Results

- Camera and screen never freeze during patient interaction.
- Kiosk response time dropped from 12.8-15.0s down to 5.0-8.0s.
- Overall system responsiveness improved by approximately 96%.



KIOSK WORKFLOW DEMONSTRATION

Subsystem Pre-Warming

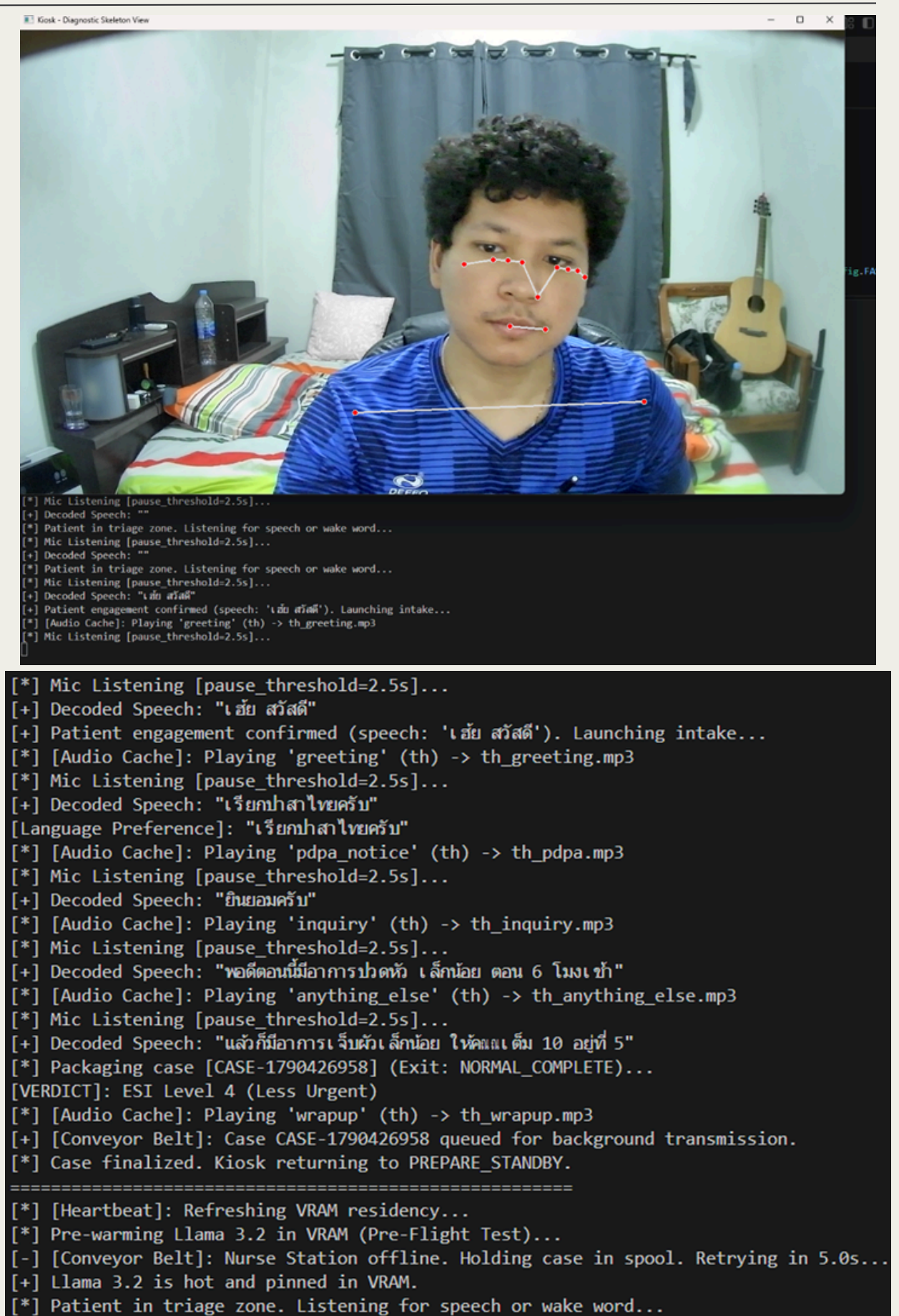
- Models load on startup so the kiosk responds instantly

Voice Controlled Process

- Activates via speech wake-word (e.g., "Hello..").
- Asks and confirms verbal PDPA consent (e.g., "I Accept.")
- Conducts clinical symptom interview and collects pain scores.

Case Verdict

- The AI classified the case with ESI Level while the background spooling system safely saved the data for nurses.



CLINICAL HANDOFF & NURSE DASHBOARD

Complete SOAP Documentation

- Subjective is patient symptoms and pain scores from voice intake.
- Objective contain Physical posture and skin vision analysis results.
- Assessment & Plan are Recommended triage level and nursing steps.

Planned Nurse Dashboard (Web-App)

- Plan to built with Node.js and an SQL database running over the local LAN so nurses can see prioritized patient cards and review snapshots.

FUTURE WORK ROADMAP

Fix Model 02 Recall & Overfitting

- Modify the threshold down to boost recall above 80% to prevent overfitting

Complete Nurse Dashboard Deployment

- Finish the web interface and connect it with hospital databases before the final defense in December.

Lip Mucosa Analysis for Pallor

- Track lip outlines using digital face points to accurately detect sudden paleness with CNN, even with makeup.

External Sensor Integration

- Add an infrared thermometer via serial port to measure exact body temperature (to collect more data)

Q & A